Physics of the Earth and Planetary Interiors, 25 (1981) 113–118
Elsevier Scientific Publishing Company, Amsterdam – Printed in The Netherlands

PALEOMAGNETIC POLES FOR THE CARBONIFEROUS BRUSH CREEK LIMESTONE AND BUFFALO SILTSTONE FROM SOUTHWESTERN PENNSYLVANIA

M.A. PAYNE [1], S.J. SHULIK, J. DONAHUE, H.B. ROLLINS and V.A. SCHMIDT
Department of Geology and Planetary Science, University of Pittsburgh, Pittsburgh, PA 15260 (U.S.A.)

(Received August 1, 1979; revised and accepted September 15, 1980)

Payne, M.A., Shulik, S.J., Donahue, J., Rollins, H.B. and Schmidt, V.A., 1981. Paleomagnetic poles for the Carboniferous Brush Creek limestone and Buffalo siltstone from southwestern Pennsylvania. Phys. Earth Planet. Inter., 25: 113–118.

We report partial results of a larger project being undertaken in the Appalachian Basin to determine the character of the geomagnetic field in the Carboniferous. The Brush Creek limestone, which contains abundant terrigenous matter, yields reliable results from three different sites, while the overlying Buffalo siltstone appears at one locality. AF demagnetization curves, IRM acquisition curves, thermomagnetic analysis, and optical reflection microscopy indicate that the remanence carriers are magnetites or titanomagnetites, which are probably primary in origin. Stepwise AF demagnetization reveals that the best demagnetizing fields are 15–20 and 20–30 mT for the limestone and siltstone units, respectively. A total of 93 specimens from 20 samples (oriented hand samples and field-drilled cores) yields a paleopole at S 36.1, W 55.7, with an alpha-95 of 4.2° and a k of 13.1 for the Brush Creek limestone. The Buffalo siltstone paleopole is located at S 27.4, W 57.0, with alpha-95 = 6.1° and k = 13.0 using 45 specimens from 31 oriented hand samples. These results agree fairly well with the mean reversed Carboniferous paleopole and Noltimier's coal pole, but are sufficiently different to raise the possibility of doing magnetostratigraphic correlations.

1. Introduction

Paleomagnetic poles have important applications in unraveling the tectonic history of a region and for magnetostratigraphic correlations. We report the results of part of a larger project being undertaken in the Appalachian Basin of the eastern United States. The project involves the investigation of the character of the paleomagnetic field during the Carboniferous with an eye to possible applications in magnetostratigraphic correlations. The major findings of the project will appear elsewhere. The two units reported here are the Brush Creek limestone and the Buffalo siltstone. They were deposited during the Upper Pennsylvanian when a shallow brackish to marine sea occupied much of the Appalachian Basin in southwestern Pennsylvania. At that time a linear belt of crystalline highlands was situated nearby to the southeast, providing a source of abundant terrigenous input to the basin. Both units appear in the stratigraphic column near the bottom of what is known locally as the Conemaugh Group. The bottom half of the Conemaugh is the Glenshaw Formation which is characterized by a sequence of four marine units (locally containing limestones), of which the Brush Creek is the lowermost. It outcrops at several localities in the area to the north of Pittsburgh. The Buffalo siltstone was found locally at only one site, but usually appears locally as the coarse-grained cut-and-fill channel or distributary bar sandstone called the Buffalo sandstone. The siltstone was deposited in a lower delta plain complex. Lithofacies changes are rapid both horizontally and vertically in the local stratigraphic column. This relationship is typical of the Appalachian Plateau province, and has in general precluded the use of traditional

[1] Present address: Department of Geoscience, New Mexico Tech, Socorro, NM 87801, U.S.A.

0031-9201/81/0000–0000/$02.50 © 1981 Elsevier Scientific Publishing Company

layer-cake geology as a stratigraphic correlation tool. We hope to use the paleomagnetic data to gain a firmer understanding of the stratigraphic relationships in this region.

2. Rock magnetic results

The Brush Creek limestone is very "dirty", containing abundant terrigenous silt and clay, and has proved to be amenable to paleomagnetic measurements. At the outcrop or in hand sample, it appears as a distinctive dark gray fossiliferous unit. Thin section analysis reveals the presence of much silt and clay. The biograins consist mainly of shell fragments. There are many angular, equant fine and silt-sized quartz grains present (0.1 mm or less). Under reflecting light one can see many equant magnetite or titanomagnetite grains, generally less than 2 μm across. There are occasional groups of magnetites or titanomagnetites which are on the order of microns across. These appear to be either large single grains or clusters of smaller individual grains; occasional oxidation rims are present, probably occurring as the result of diagenesis.

Rock magnetic experiments have been performed on the Brush Creek limestone. A preliminary thermomagnetic analysis done in an equilibrium atmosphere indicates the presence of a range of titanomagnetites along the magnetite–ulvospinel join. When subjected to increasingly higher direct current magnetic fields up to 900 mT, the samples saturate in the range 150–300 mT. The median destructive demagnetizing field for NRM for 21 specimens is 17.7 ± 8.1 mT, while for IRM for 11 specimens it is 20.5 ± 3.2 mT. These tests indicate that the remanence carriers are probably magnetites or titanomagnetites, which we believe to be primary in origin, becoming aligned in the Carboniferous geomagnetic field at or shortly after deposition in the Appalachian Basin.

The Buffalo siltstone appears as a brown–gray, very fissile, platy unit. In thin section it shows a matrix of clays, organic detritus and quartz, surrounding larger angular, equant quartz grains of fine-sand and silt size. The samples saturate in the range 200–500 mT when subjected to increasingly higher magnetic fields up to a maximum of 900 mT. The median destructive field for NRM for 43 specimens is 21.1 ± 4.8 mT, while for IRM for four specimens it is 22.3 ± 0.5 mT. As with the Brush Creek limestone, we believe the remanence carriers for the Buffalo siltstone to be primary magnetites or titanomagnetites which were aligned in the Carboniferous field at or near the time of deposition.

3. Paleomagnetic results

A total of six field-oriented hand samples and sixteen drilled cores were collected within the Brush Creek limestone unit from five separate sites. The sites included Sewickley Bridge (site 2 in Fig. 1), Glenfield I-79 (site 3), Avonworth Park (site 4), Glenshaw Glass (site 8), and Pennsylvania Gameland No. 177 (site 10), in a traverse from west to east just north of Pittsburgh. A distance of 21 km separates Sewickley Bridge from Pennsylvania Gameland. The thickness varies from 45 cm at Sewickley Bridge to 15 cm at Pennsylvania Gameland. In all cases, the hand samples represented the entire thickness of the unit at that locality. The hand samples were oriented in the field using a rotatable platform mounted on a tripod. They were then returned to the laboratory and cored on a drill press with the field-oriented reference plane mounted horizontally. One block (from Glenshaw Glass) was reoriented in the laboratory before drilling so that a more convenient plane than the field-oriented one was horizontal. Conversion from sample coordinates back to field coordinates was accomplished using an orthogonal transformation (Schmidt, 1974). The samples which were obtained by drilling in the field were oriented using a rotatable platform mounted on a collar that fitted over the core at the outcrop. All of the samples were 2.33 cm in diameter. They were sliced to a height of 2.35 cm. All of the samples were measured in an ScT cryogenic magnetometer. They were demagnetized in a two-axis tumbler system enclosed within a three-stage mumetal shield. The peak alternating field (AF) attainable in our system is about 140 mT.

Approximately one-fourth of the specimens from each locality was subjected to stepwise AF demagnetization up to a maximum peak field of 80 mT. A Zijderveld (1967) diagram was then constructed for each specimen. Figure 2a is such a diagram for a typical reliable specimen, this one from the western-most

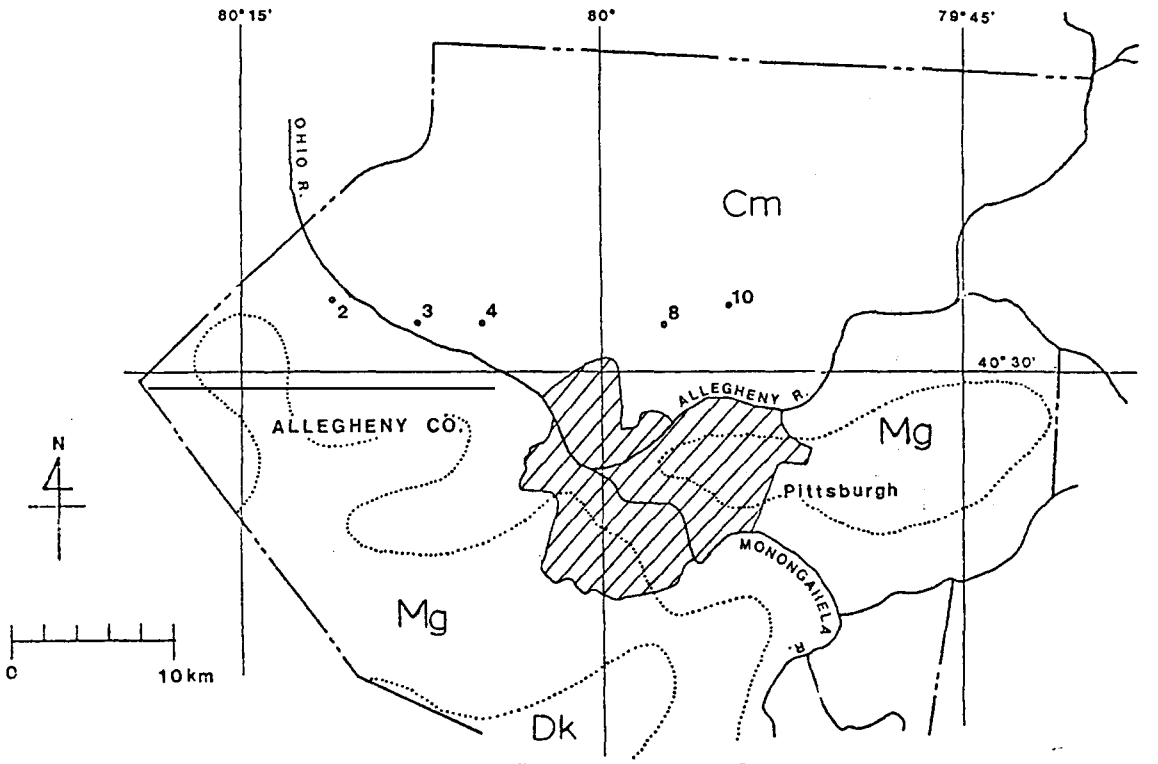
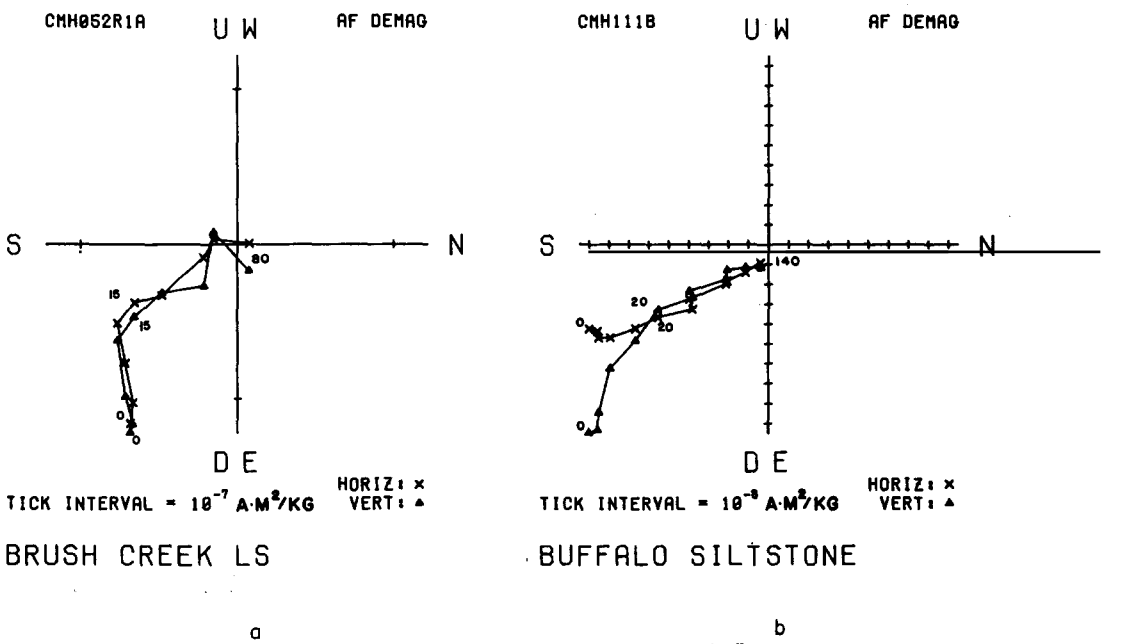


Fig. 1. Sampling sites for samples involved in this study. The Brush Creek limestone was collected at all five sites shown, while the Buffalo siltstone samples were collected at site 2. The contacts between geologic boundaries (dotted lines) are approximate. The following groups are included: Conemaugh (Cm), Monongahela (Mg), and Dunkard (Dk). All but the Dunkard are Pennsylvanian (Upper Carboniferous).



a b

Fig. 2. Typical Zijderveld diagrams. a: Brush Creek limestone. A direction intermediate between the Carboniferous and present-day fields is removed up to 10 mT, then the reversed Carboniferous field is removed. b: Buffalo siltstone. An intermediate direction is removed up to about 20 mT, then the reversed Carboniferous field is removed.

locality, Sewickley Bridge. Up to 10 mT, a direction intermediate between the reversed Carboniferous field and the present-day field is removed. Above that, the vector declines roughly in a straight line along the reversed Carboniferous direction until it nearly reaches the origin at 80 mT. Analyses such as this revealed that the best blanket demagnetizing field for the Sewickley Bridge and Glenshaw Glass specimens was 15 mT, while for Pennsylvania Gameland it was 20 mT. The Avonworth Park and Glenfield I-79 specimens from one hand sample each yielded unreliable results, and appear to have been heavily overprinted since the Cretaceous Period. These two were more visibly weathered than those samples discussed earlier. A total of 93 specimens from the remaining 20 samples were used from the other three localities. The virtual geomagnetic poles obtained at a demagnetization level of 15 or 20 mT (as discussed earlier) are plotted in Fig. 3a, together with the mean paleopole for the unit and the oval of confidence. The paleopole is located at latitude S 36.1, longitude W 55.7. The alpha-95 is 4.2° with a kappa of 13.1. The mean scalar intensity is 8.37×10^{-8} A-m^2 kg^{-1}.

Two drilled cores and 29 field-oriented hand samples represent the collection of the fissile Buffalo siltstone from the Sewickley Bridge site (2 on Fig. 1). The samples span a range from 50 to 600 cm above the top of the 45 cm thick limestone unit. The cored samples measured 2.33 cm in diameter and 2.35 cm in height. They were oriented in the field in the same manner as the cored limestone samples. The hand samples were oriented in the field by placing a Brunton compass on a flat part of the rock because they were in general too small to be oriented using a tripod. This method is more susceptible to errors than that using a tripod, but seemed to work well. Some of the samples were reoriented in the laboratory in the same way as the limestone block. The hand samples were cored on the drill press. They measured 3.35 cm in diameter with heights varying from about 1 cm to 3.4 cm. All of the samples were measured in an ScT cryogenic magnetometer. They were demagnetized in the apparatus described earlier.

As with the Brush Creek limestone, approximately one-fourth of the specimens were subjected to stepwise AF demagnetization up to a peak field of 80 mT. The Zijderveld diagram in Fig. 2b is a typical example of the results. The magnetization vector has a little

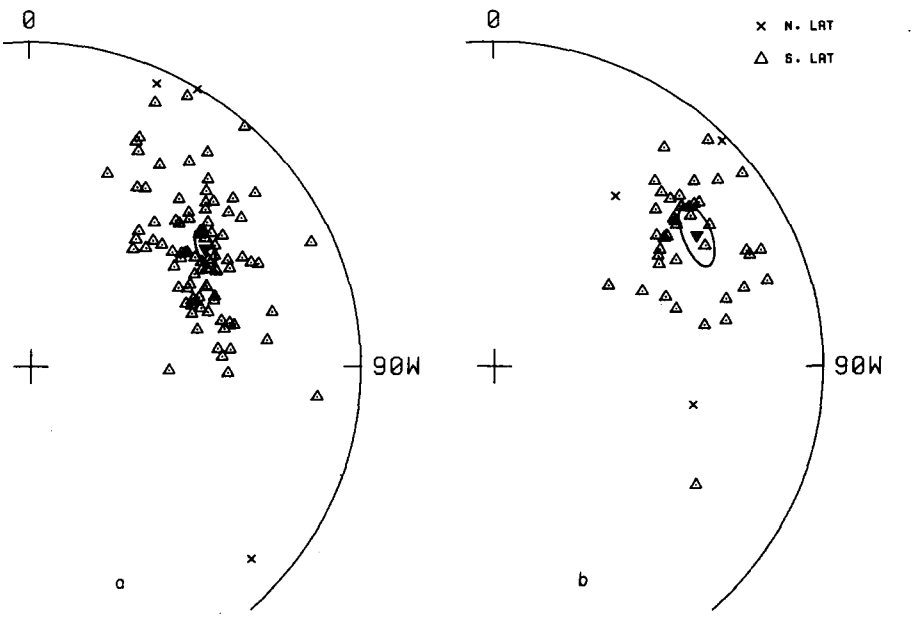


Fig. 3. Specimen VGPs and unit mean paleopole for (a) Brush Creek limestone at the 15 or 20 mT demagnetization level and (b) Buffalo siltstone at the 20 mT demagnetization level. The mean paleopoles are shown as inverted solid triangles. The oval of confidence is included for each unit.

contamination in the lower coercivity components, then generally moves in a straight line along the reversed Carboniferous direction above 10 to 20 mT to the vicinity of the origin at 80 mT. The remaining three-fourths of the specimens were then demagnetized at peak fields of 20, 25, and 30 mT. The paleopoles from each of these three demagnetization levels are virtually identical, and indeed, are identical to the vector removed over the interval from 20 to 30 mT. Figure 3b shows the virtual geomagnetic poles of the 45 specimens from the site, together with the site mean paleopole and the oval of confidence from the 20 mT demagnetization level. The pole is at latitude S 27.4, longitude W 57.0. The alpha-95 is 6.1° with a kappa of 13.0. The mean scalar intensity is 8.17×10^{-8} A-m^2 kg^{-1}.

4. Discussion

Figure 4 shows the position of the two paleopoles obtained from this study, as well as the mean Carboniferous pole from North America (McElhinny, 1973, p. 202) and the coal pole from a study of coals in Ohio, Kentucky, and West Virginia (Kopacz and Noltimier, 1976). In each case, the normal Carboniferous pole is shown on the equal-area projection. The mean North American pole is included within the oval of

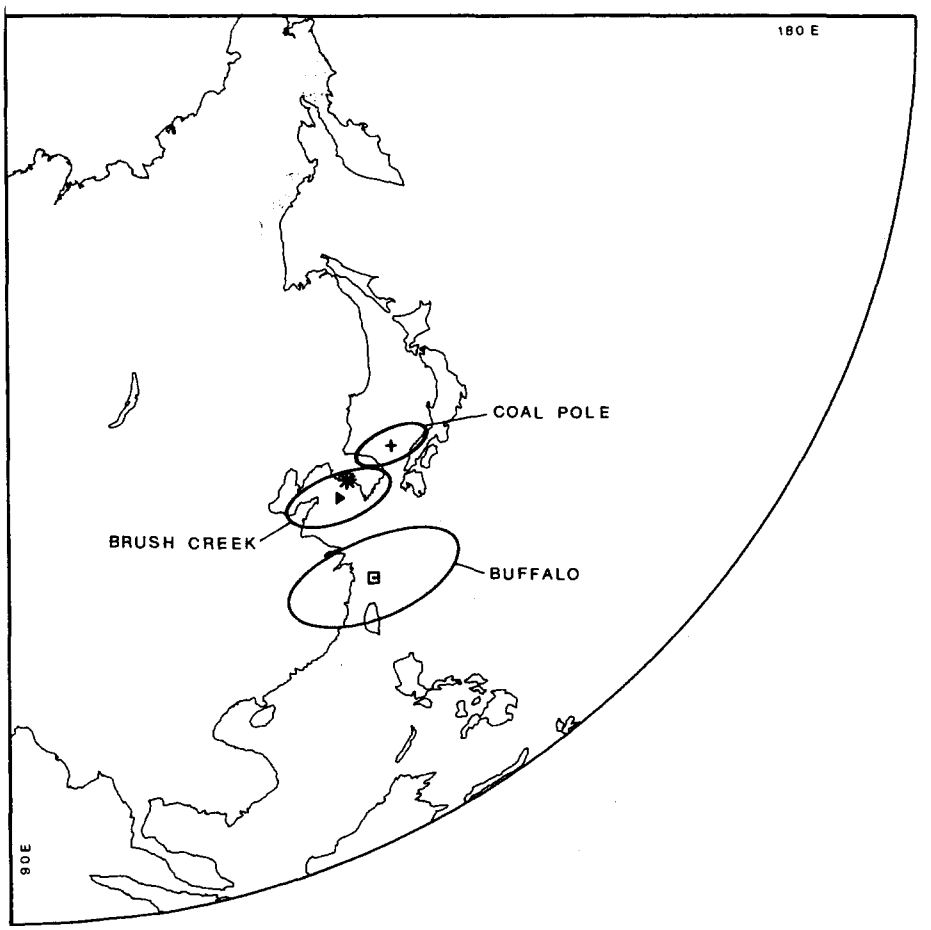


Fig. 4. Paleopoles for Brush Creek limestone (triangle) and Buffalo siltstone (square) from this study; the mean Carboniferous pole (asterisk) from McElhinny (1973); and the coal pole (plus) from Kopacz and Noltimier (1976). The poles from this study, while actually reversed, are shown as normal.

confidence of the Brush Creek limestone unit and the three ovals of confidence marginally do not overlap. The first result may be fortuitous, since ten of the thirteen poles used by McElhinny (1973) come from studies done in the New England–Canadian Maritime region. Kent and Opdyke (1978), as a consequence of research on the Devonian Catskill red beds in New York state, have proposed that this region was separate from cratonic North America, of which the Appalachian Basin is a part, until the Upper Carboniferous. Alternatively, the New England–Canadian Maritime region may have been connected to cratonic North America earlier than that, perhaps during the Lower Carboniferous.

The fact that the ovals of confidence do not overlap has possible implications for magnetostratigraphic correlations. Since so many samples are involved, it would seem that the divergence of the paleopoles is due to some sort of magnetizing process or processes acting differently on the three distinct lithologies or that the three poles are truly distinct. An inclination error seems unlikely since the Buffalo siltstone, with distinct bedding, has a steeper, rather than shallower, inclination than the massive Brush Creek limestone. If the poles are truly distinct, then secular variation may not be completely averaged out during deposition of a unit. Rollins and Donahue (1975) lend credence to this hypothesis by suggesting that deposition of the Brush Creek limestone may have occurred over a time span of a few thousands of years, and the Buffalo siltstone was certainly deposited over a much smaller time interval. Indeed, data to be published elsewhere indicate that the general trend of secular oscillations of the geomagnetic field may be recorded in the Brush Creek limestone. Turner (1975) obtained similar results from a Carboniferous limestone in England. Since the ages of the poles recorded in order of decreasing age are coal pole, Brush Creek limestone, and Buffalo siltstone, a kind of polar wandering path for a short time interval of the Upper Carboniferous may be indicated. This could possibly be used to do magnetostratigraphic correlations over short distances such as within the Appalachian Basin, where normal stratigraphic correlation techniques have proved to be difficult to apply.

Acknowledgments

We thank Bruce Cain for his help in sample collecting. This work was supported by NSF Grant EAR 7612470.

References

Kent, D.V. and Opdyke, N.D., 1978. Paleomagnetism of the Devonian Catskill red beds: evidence for motion of the coastal New England – Canadian Maritime region relative to cratonic North America, J. Geophys. Res., 83: 4441–4450.

Kopacz, M.A. and Noltimier, H.C., 1976. The coal pole and the Kittanning reversal. Eos Trans. Am. Geophys. Union, 57: 238 (abstract).

McElhinny, M.W., 1973. Paleomagnetism and Plate Tectonics. Cambridge University Press, London, 358 pp.

Rollins, H.B. and Donahue, J., 1975. Towards a theoretical basis of paleoecology: concepts of community dynamics. Lethaia, 8: 255–270.

Schmidt, V.A., 1974. On the use of orthogonal transformations in the reduction of paleomagnetic data. J. Geomagn. Geoelectr., 26: 475–486.

Turner, P., 1975. Paleozoic secular variation recorded in Pendleside limestone. Nature (London), 257: 207–208.

Zijderveld, J.D.A., 1967. AC demagnetization of rocks: analysis of results. In: D.W. Collinson et al. (Editors), Methods in Paleomagnetism, Elsevier, Amsterdam, pp. 254–286.